Sheikh Shihab Uddin

AI Engineering Internship Candidate | *Agentic AI Systems* | *LLM Tool Calling & MCP*

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PROFESSIONAL SUMMARY

Computer Science & Engineering student at BRAC University who built a multi-provider LLM agent harness with MCP tool integrations, a locked-down Docker code sandbox, Telegram human-approval gates, and a streaming web interface. Strong in Python, security-minded design, and agent safety, with embedded robotics experience and video skills for clear technical demos. Seeking an AI engineering internship to build reliable, secure agent systems.

EDUCATION

B.Sc. in Computer Science and Engineering | *BRAC University, Dhaka* *Expected 2028*
Relevant coursework: Data Structures & Algorithms, Probability & Statistics, Advanced Calculus

EXPERIENCE & LEADERSHIP

STEM & CS Tutor | *Dhaka* *Jan 2023 – Present*

- Taught 10+ SSC, HSC, and first-year university students in mathematics, physics, ICT, and computer science, turning topics like calculus and algorithm execution into structured lessons.

Active Member | *BRAC University Robotics Club (BURO)* *Oct 2026 – Present*

- Worked with hardware and software teams on embedded systems, sensor integration, and robotics prototypes.

PROJECTS

Autonomous Multi-LLM Agent Harness with MCP & Sandboxed Execution | GitHub
Python, TypeScript, Next.js, Docker, Ollama, MCP, Telegram Bot API

- **Multi-LLM Orchestration:** Built a local-first AI agent harness with a streaming tool-calling loop and persistent session state, with automatic fallback across 5 LLM providers (Ollama, Gemini, Groq, OpenAI, Anthropic).
- **MCP Tool Integration:** Integrated 7 MCP servers (Time, Web Fetch, Memory, GitHub, Context7, Playwright, Sequential Thinking) with schema validation and isolated error handling so one server failure never blocks the agent.
- **Secure Code Execution:** Engineered a Docker sandbox for LLM-generated Python: no network, read-only filesystem, 0.5 CPU / 256 MiB limits, and a 30-second timeout.
- **Human-in-the-Loop Safety:** Implemented a Telegram approval gate with Approve/Deny buttons and a 60-second auto-timeout for any data-mutating action.
- **Web Interface:** Built a Next.js/TypeScript chat UI with live token streaming (SSE), a tool-activity log, and skill/MCP management; secured with bearer-token auth and strict CORS origins, and mutating tools are denied in web mode.
- **Agent Evaluation:** Created an offline evaluation and unit-test suite of 84 tests (14 CLI-specific) covering tool-selection accuracy, context retrieval efficiency, and failure recovery.

Encrypted Credential Vault & Password Manager | GitHub
Python, Cryptography

- Built a CLI password manager with symmetric encryption at rest, modular architecture, .env-based secret isolation, and documented secure API-key setup.

Autonomous Robotic Watercraft *Oct 2026*
Arduino, Embedded C/C++, Sensors

- Built an autonomous boat with waypoint navigation set remotely, plus sensor data collection, using Arduino and motor drivers.
- Implemented motor-control logic and hardware-software communication for real-time sensor processing.

TECHNICAL SKILLS

Agentic AI: LLM Agents, Tool Use / Function Calling, Model Context Protocol (MCP) Integration, ReAct Loops, Multi-LLM Orchestration, Human-in-the-Loop (HITL) Workflows, Agent Evaluation, Context Management
LLM Providers: OpenAI, Anthropic, Google Gemini, Groq, Ollama (local models)
Languages & Web: Python (Asyncio, Pydantic, Aiohttp), TypeScript/JavaScript, React, Next.js, Tailwind CSS, Java, C/C++
Infrastructure & Security: Docker, Linux/Bash, Git/GitHub, uv, Sandboxed Code Execution, Secrets & Environment Management
Testing: unittest, Mocking, Offline Regression Testing, Eval Benchmarking
Foundations: Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Calculus

ADDITIONAL

Creative & Multimodal: 3D motion graphics, visual storytelling, and technical video editing; used to produce demo videos and architecture walkthroughs for agent projects. Portfolio
Languages: Bangla (Native), English (Professional Working Proficiency)